MODEL IDENTIFICATION AND DATA ANALYSIS – Part 1 (Automation&Control Engineering), A.Y. 2021/2022

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Surname	Name	University ID Number	Signature
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- Write the solutions in the blank areas (use the back of the page, if needed)
- The number of pages is 4. Additional papers will not be considered.
- Clarity, order and precision are strongly considered for the final evaluation.

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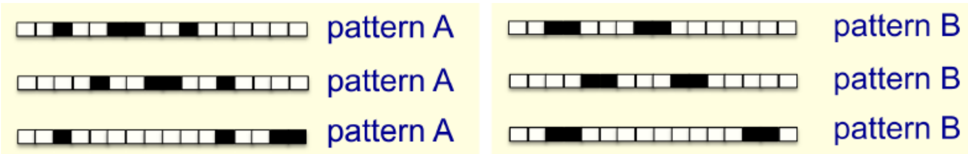
1. [Neural networks]

Consider the binary classification problem of mapping a sequence of black/white points onto two different patters (pattern A or B). See, e.g., the figure below.



You are provided with a dataset for training and the information that it is *impossible* for a linear classifier to solve the problem. Specifically, the training set consists of patterns A and B in all possible translations, with wraparound.

a. Consider now a neural network (NN) made of an input layer with a linear activation function, followed by a linear hidden layer with a logistic output. Can such an architecture perfectly classify all of the training examples? Why or why not?
b. Describe the architecture of a feed-forward NN.
c. Explain why using a feed-forward NN with a sigmoidal activation function like $tanh$ may be preferred to the multi-layer perceptron.

Solutions

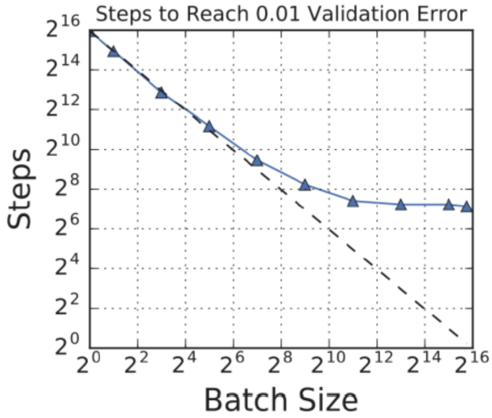
a. No. All layers but the output one (used for classification) are linear, and any composition of linear layers is still linear.
b. See the lecture notes (chapter 09).
c. See the lecture notes (chapter 09).

2. [Gradient descent]

a. Describe the classical *gradient descent* approach to optimization and its stochastic counterpart.

b. Why is gradient descent employed for training logistic models instead of ordinary least squares?

c. Consider the following plot of the number of stochastic gradient descent iterations required to reach a given loss, as a function of the batch size (namely, the number of data used to compute the gradient at each iteration). For small batch sizes, the number of iterations required to reach the target loss decreases as the batch size increases. Why is that?



d. In the same plot, it can be observed that, for large batch sizes, the number of iterations does not change much as the batch size is increased. Why is that?

Solutions

a. See the lecture notes (chapter 04).

b. See the lecture notes (chapter 04).

c. Larger batch sizes reduces the variance in the gradient estimation of SGD. Hence, larger batch converges faster than smaller batch.

d. As the batch size grows larger, SGD becomes practically equivalent to full-batch gradient descent.

3. [Learning theory]

a. Discuss the *bias-variance trade-off* in linear regression.
b. Is it possible to distinguish between the bias and variance errors looking at the learning curves? If yes, how such curves could then be used for model selection?
c. Which degree of freedom of the learning problem can be varied to decrease the bias error? And the variance? Why?

Solutions

a. See the lecture notes (chapter 07).
b. See the lecture notes (chapter 08).
c. Bias: hypothesis set; variance: number of samples. See the lecture notes (chapter 08).

4. [Stochastic processes]

Give (at least) one example of a non-stationary stochastic process. Why the sample estimators of mean and covariance function studied in this course *cannot* be used to estimate the statistical mean and the real covariance function of such a process?

Solutions

See the lecture notes (chapter 13).